
Software Requirement Specification

Waste Management & Recycling Tracking System

Organization: Donato Impex Limited

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1. Executive Summary

This document outlines the Waste Management & Recycling Tracking System designed for Donato Impex Limited. The system provides a complete solution to track material from receipt through processing stages to final inventory, calculate costs accurately, manage finances, and generate comprehensive reports.

The system helps streamline operations, reduce losses through better visibility, and maintain accurate financial records for decision-making and compliance.

2. Organization Overview

Company Name: Donato Impex Limited

Business Focus: Sustainable waste management and recycling solutions

Products:

- PP Injection materials
- HDPE Blow materials
- LD (Low-Density) materials
- Rafia Pellets

3. System Purpose

The system addresses the following needs:

- Track material from receipt to final product
- Calculate actual costs considering transport losses
- Monitor material losses at each processing stage
- Maintain organized inventory
- Record and manage all financial transactions
- Generate reports for management decision-making
- Maintain audit trails and documentation
- Provide real-time visibility into operations

4. User Roles

Admin

- Complete system access
- User management
- System configuration
- Financial oversight
- Report generation

Manager/Supervisor

- Monitor operations
- Review reports
- Approve activities
- Oversee staff work
- Track performance

Operator/Staff

- Enter material receipt details
- Record processing stage completions
- Update inventory

- Record expenses and income
 - Upload supporting documents
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5. Features Overview

5.1 User Authentication & Access Control

- Secure login with username and password
 - Role-based access (Admin, Manager, Operator)
 - User account management
 - Activity tracking and logs
 - Session management
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5.2 Material Receipt Module

Record incoming raw materials with:

- Material type selection
- Quantity received (in KG)
- Quantity sent by supplier
- Purchase price and total cost
- Supplier/Sender details
- Receipt date and time

Automatic Calculations:

- Transport loss quantity (in KG)
- Transport loss cost (in ₹)
- Effective cost per KG after losses
- Example: 100 KG sent at ₹20/KG, received 80 KG → System calculates effective cost and loss impact

Additional Features:

- Photo upload for documentation
 - Supplier information storage
 - Material batch creation with unique ID
 - Approval workflow
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5.3 Processing Workflow Module

Track material through 4 processing stages:

Stage 1: Sorting

- Input material quantity
- Output quantity after sorting
- Automatic loss calculation
- Quality grading (Grade A, B, C, Rejected)
- Operator details and notes

Stage 2: Crushing

- Input from sorting output
- Output quantity after crushing
- Loss calculation
- Processing notes

Stage 3: Washing

- Input from crushing

-
- Output after washing
 - Loss tracking
 - Processing details

Stage 4: Pelletizing

- Input from washing
- Final product quantity
- Loss at final stage
- Finished product specifications

Loss Tracking Features:

- Automatic loss calculation at each stage (in KG and %)
- Loss cost calculation based on input material cost
- Cumulative loss tracking (from receipt to finished product)
- Loss history and analysis

5.4 Sorting Module

Detailed interface for the sorting stage:

- Categorize mixed raw materials
- Separate into different quality grades
- Track rejected/defective materials
- Document sorting process with photos
- Record operator and timing details

5.5 Inventory Management

Finished Goods Inventory:

- Product name and code
- Current stock quantity
- Unit type (KG, Pieces, Bags, etc.)
- Unit price
- Stock status tracking

Inventory Operations:

- Automatic stock additions from processing completion
- Manual stock deductions
- Stock level adjustments with approval
- Stock history tracking

Stock Monitoring:

- Low stock alerts when inventory below reorder level
- Critical stock notifications
- Empty stock alerts
- Reorder reminders

Additional Features:

- Stock value calculation
 - Product-wise inventory summary
 - Inventory status reports
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5.6 Dashboard & Analytics

Main Dashboard Shows:

- Total material received (current month/year-to-date)
- Material processed by stage
- Finished products produced
- Current inventory value
- Monthly income and expenses
- Balance summary

Key Visualizations:

Loss Per Material Graph:

- Visual comparison of losses at each processing stage
- Shows loss quantity by material type
- Identifies highest loss stages
- Helps target improvement areas

Material Flow Chart:

- Shows path from input to finished product
- Displays quantity at each stage
- Highlights losses at each step

Inventory Status:

- Stock levels by product
- Low stock indicators
- Inventory value breakdown

Financial Summary:

- Daily/monthly income
- Daily/monthly expenses
- Profit/loss overview
- Cash position

5.7 Income & Expense Management

Income Recording:

- Sales transaction entry
- Customer details
- Product sold and quantity
- Selling price and amount
- Invoice generation
- Payment details and date

Expense Recording:

- Expense type/category
- Amount spent
- Payment channel (Cash, GPay, Bank Transfer, Cheque)
- Date of expense
- Description and notes
- Document upload (receipts, bills)

Expense Categories:

- Material costs
- Transportation

- Utilities (electricity, water)
 - Labor and wages
 - Equipment maintenance
 - Administrative costs
 - Other expenses
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5.8 Financial Reports

Balance Sheet:

- Assets (Cash, Bank Balance, Inventory Value)
- Liabilities (if any)
- Equity overview
- Complete financial position

Profit & Loss Statement:

- Total revenue
- Total expenses
- Gross profit
- Net profit/loss
- Profit margins

Cash Flow Report:

- Cash inflows (from sales, loans)
- Cash outflows (expenses, purchases)
- Net cash flow
- Ending cash position

Bank Account Management:

- Multiple bank account tracking
 - Bank balance by account
 - Transaction history
 - Account summary
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5.9 Loss Analysis & Reporting

Loss Tracking Shows:

- Total loss by processing stage (in KG and %)
- Loss by material type
- Cost impact of each loss
- Cumulative loss from input to output
- Loss trends over time

Loss Reports Include:

- Monthly loss summary
- Loss by material type
- Loss percentage by stage
- Trend analysis (improving or declining)
- Comparison with previous periods

Loss Insights:

- Identifies highest loss stages
- Highlights problem areas
- Enables process improvement focus

5.10 Batch Tracking

- Each material receipt gets unique batch ID
- Track batch through all processing stages
- View complete batch history
- Source to finished product traceability
- Batch status tracking

5.11 Quality Management

- Grade assignment (A, B, C, Rejected)
- Quality metrics tracking
- Defect documentation
- Quality trends monitoring

5.12 Document Management

- Centralized storage for all documents
- Travel documents and receipts
- Process documentation
- Transaction proofs
- Easy search and retrieval

5.13 Data Export & Reporting

Export Options:

- Export balance sheet to Excel
- Export income statement to Excel
- Export expense reports to Excel
- Export inventory reports
- Export loss analysis reports
- Export custom date-range reports

Export Features:

- Excel format for easy use
- CSV format for data analysis
- PDF format for sharing
- Scheduled automatic exports
- Email reports directly

5.14 Alerts & Notifications

- Low stock alerts
- Critical inventory warnings
- Approval request notifications
- Daily summary emails
- Important updates in real-time

5.15 Audit Trail

- Complete record of all activities
- Who did what and when

- Change history tracking
 - Ensures accountability
 - Supports compliance requirements
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6. Implementation Timeline

Total Implementation Period: 6 weeks (1.5 months)

Week 1-2: Foundation Setup

- System setup and configuration
- User accounts and role creation
- Team training begins
- Dashboard setup

Week 3-4: Core Operational Setup

- Material receipt module implementation
- Processing stages configuration
- Inventory system setup
- Sorting module setup

Week 5-6: Financial & Reporting

- Income and expense system setup
- Balance sheet and reports configuration
- Data export setup
- Dashboard completion and testing
- Final training and deployment

Throughout Implementation:

- Daily support and coordination
 - Weekly status meetings
 - User feedback incorporation
 - System testing and refinement
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7. Training & Support

Training Provided

Admin Training (1.5 days):

- System configuration
- User management
- Report generation
- Financial module setup
- **System maintenance**

Manager Training (1 day):

- Dashboard usage
- Report interpretation
- Processing oversight
- Financial review

Operator Training (0.5 day):

- Material receipt entry
- Processing data input
- Expense recording
- Income recording

- Photo upload and documentation

Training Materials:

- Video tutorials
- User manuals
- Quick reference guides
- FAQ document
- Sample data for practice

Support Services**During Implementation (6 weeks):**

- Dedicated support team
- Daily availability
- Quick issue resolution

After Launch (First 30 days):

- Intensive support
- Daily check-ins
- User assistance
- Issue resolution

Ongoing Support:

- 24/7 helpdesk available
- Email and phone support
- Monthly system reviews
- Regular feature updates
- Security updates as needed

8. Key Benefits

- ✓ Complete Visibility - See all material from receipt to finished product
- ✓ Accurate Cost Tracking - Know exact costs including transport losses
- ✓ Loss Monitoring - Identify and track losses at each processing stage
- ✓ Automated Calculations - Reduce manual errors
- ✓ Real-Time Reports - Instant access to financial and operational data
- ✓ Better Decisions - Data-driven insights for management
- ✓ Organized Records - All documents and transactions in one place
- ✓ Easy to Use - Simple, intuitive interface
- ✓ Audit Ready - Complete trails for compliance
- ✓ Scalable - Grows with your business

9. System Access

The system will be accessible through:

- Web browser from office and field
 - Mobile-friendly interface
 - Internet-based (cloud) for reliability
 - Secure login required
 - Data accessible anytime, anywhere with internet
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10. Data Security

- Secure username/password login
 - Role-based access control
 - Automatic data backup daily
 - Data encryption for protection
 - Activity logging for accountability
 - Secure data storage
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11. Terms & Conditions

Scope

- All features as listed in this document
- Implementation over 6 weeks
- Training for all user roles
- 30 days post-launch support included

Modifications

- Scope changes require separate evaluation
- Additional features may incur extra cost/timeline
- Changes require written approval

Timeline

- 6-week implementation is subject to:
 - Timely client feedback
 - Team availability for training
 - No major requirement changes

Responsibilities

Our Team Will:

- Deliver all features as specified
- Provide complete training
- Offer technical support
- Maintain system security
- Provide regular updates

Your Team Will:

- Designate key persons for coordination
- Provide timely feedback
- Ensure staff participation in training
- Provide necessary business information

Data Ownership

- All data entered remains your company's property
- You can export data anytime
- Full control over your information

Support & Warranty

- 30-day warranty on software bugs
- Free bug fixes during warranty
- Support team available for questions
- Regular system maintenance included

Confidentiality

- Both parties maintain confidentiality

- Business information kept secure
 - Standard non-disclosure practices followed
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12. Success Factors

For successful implementation:

1. Clear Leadership Support - Management commitment to system adoption
 2. Data Accuracy - Quality information entry from day one
 3. Team Training - All users complete training programs
 4. Regular Usage - Consistent system usage from start
 5. Feedback Communication - Share issues and suggestions promptly
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13. Next Steps

Week 1: Review & Planning

1. Review this document with your team
2. Identify any additional needs
3. Confirm key contacts and team members

Week 2: Kick-Off Meeting

1. Schedule demo of sample system
2. Finalize requirements
3. Agree on timeline and team assignments
4. Discuss data preparation needs

Week 3: Implementation Begins

1. Sign agreement
2. System setup starts
3. Team training begins
4. Data preparation commences

Weeks 4-8: Active Implementation

1. Features implemented phase by phase
2. Weekly status meetings
3. User testing and feedback
4. Team training continues

Week 9-10: Launch Preparation

1. Final testing and refinement
2. Staff training completion
3. Data migration finalization
4. Go-live preparation

Week 11 Onwards: Live System

1. System goes live
 2. Intensive support provided
 3. Staff assists with queries
 4. Monitor and optimize
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14. Contact Information

For questions or to begin implementation:

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15. Document Approval

This document represents the complete feature set and implementation approach for Donato Impex Limited's waste management and recycling system.

Document Status: Ready for Review and Implementation

Last Updated: October 11, 2025